

APS110 – Lecture 30

See hand written notes for drawings.

Thermodynamics

- $\text{Na}_{(s)} + \frac{1}{2} \text{Cl}_{2(g)} \rightarrow \text{NaCl}_{(s)}$
 - o At room temp, the product of this reaction is solid salt and is stable
 - o It won't spontaneously decompose into solid sodium and chlorine gas
- Ex. $\text{C}_{\text{graphite}} \rightarrow \text{C}_{\text{diamond}}$
 - o This reaction is NOT stable at room temperature (your pencil won't turn to diamond)
 - o The reaction is not going to proceed at room temperature and pressure
 - o What about the reverse reaction?
 - Technically diamond is NOT stable at room temp and pressure
 - The decomposition (reverse rxn) will occur, but very slowly
 - Because this decomp. happens SO slowly, we say that diamond is **METASTABLE**
 - Thermodynamically unstable, but kinetics of the transformation are slow

Energetics of Reactions

- Ex. $\text{Na}_{(s)} + \frac{1}{2} \text{Cl}_{2(g)} \rightarrow \text{NaCl}_{(s)}$
 - o (1) $\text{Na}_{(s)} \rightarrow \text{Na}_{(g)}$ *melting + vaporization $\rightarrow$ sublimation*
 - HEAT required $\rightarrow$ enthalpy of sublimation
 - $\Delta H_{\text{sublimation}} = 109 \text{ kJ/mol}$
 - To take a piece of solid Na and transform it into the gas phase, you need to put heat in
 - o (2) $\text{Na}_{(g)} \rightarrow \text{Na}^+ + \text{e}^-$
 - $\Delta H_{\text{ionization}} = 497 \text{ kJ/mol}$ (quite a lot)
 - o (3) $\frac{1}{2} \text{Cl}_{2(g)} \rightarrow \text{Cl}_{(g)}$
 - Chlorine free radical, unbonded $\rightarrow$ **bond dissociation** (breaking the bond in Cl_2)
 - Bond dissociation energy = 121 kJ/mol
 - o (4) $\text{Cl}_{(g)} + \text{e}^- \rightarrow \text{Cl}^-$
 - Electron affinity = -364 kJ/mol
 - Recall, Cl is a very EN atom and attracts electrons very strongly
 - Thus, this step of the reaction releases energy $\rightarrow$ exothermic
 - o (5) $\text{Na}^+ + \text{Cl}^- \rightarrow \text{NaCl}$
 - Energy associated with forming all those bonds in the crystal
 - $E_{\text{crystallization}} = -777 \text{ kJ/mol}$
 - Strongly exothermic!
 - o What is the **total** enthalpy change for the whole reaction? $\rightarrow -414 \text{ kJ/mol}$
 - Thus, the entire reaction is exothermic

- o Does it matter that solid (end product) NaCl is stable? Is every reaction that will proceed on its own exothermic?
 - Being exothermic is NOT a requirement for a reaction to proceed
 - Think of solid ice at room temperature → it will melt, but we are putting in heat (energy) for the melting to occur
 - Not Exothermic!
 - **Reactions need NOT be exothermic to proceed on their own (spontaneous)**

DEMO

- Test tubes filled with H₂ (hydrogen gas) and Cl₂ (chlorine gas)
- Can we get a reaction started?
 - o Cl_(g) → 2 Cl• *bond dissociation*
 - o How much energy do we need to put in for this rxn to occur?
 - o Bond dissociation energy for Cl₂ = 2.57 eV
 - o Recall, visible light is 2-3 eV
 - So we can use a wavelength of visible light to break this chlorine bond
 - o Bond dissociation energy for H₂ = 4.52 eV
 - No wavelength of visible light could break the bond in Hydrogen gas (would need to go into UV light)
 - o Cl• free radical is VERY reactive → will attack H₂
 - Cl• + H₂ → HCl + H•
 - H• + Cl₂ → HCl + Cl•
 - Reaction will propagate through sample
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 - o E = hc/λ
 - Red: 635nm → 1.9 eV
 - Green: 520nm → 2.39 eV
 - Blue: 445nm → 2.79 eV
 - o NOTE: Cl₂ has a green colour (makes sense with bond dissociation energy)
 - Blue photons are being absorbed
- Different intensities of lasers (various wattages)
 - o Is this important? Intensity of the light?
 - o We need the **photon energy** to be high enough → green laser has high intensity but not enough photon energy
 - o The blue laser is the only one with enough photon energy to initiate the reaction
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